

Davis Rattनाविजै

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ABOUT ME

Third-year Mechanical Engineering student interested in supply chain, operations, and mechanical engineering. I bring technical experience through Formula SAE and manufacturing, complemented by experience in customer service and entrepreneurship.

EDUCATION

Cornell University, Duffield College of Engineering, Ithaca, New York Expected May 2028
Bachelor of Science in Mechanical Engineering GPA: 3.73

EXPERIENCE

Machine Shop Staff Feb 2026 - Present

Manufacturing Learning Studio, Cornell University, Ithaca, NY

- Operate manual mills, lathes, 3D scanners, and Haas CNC equipment for manufacturing, inspection, and maintenance
- Train engineering students in machine operation, tool selection, tolerance analysis, and efficient machining practices
- Develop technical manuals, safety documentation, and instructional materials to standardize equipment operation

Suspension Lead Jun 2026 - Present

Cornell Formula SAE Racing, Cornell University, Ithaca, NY

- Lead a 6-person suspension team through design, manufacturing, and integration, overseeing CAD and engineering analysis
- Coordinate suspension integration across chassis, steering, tires, and aerodynamics, balancing VD targets with suspension
- Perform kinematic and vehicle dynamics simulations to optimize suspension geometry and define rates, stiffnesses, and design loads using vehicle driving data and driver feedback
- Develop physical tests to validate suspension and vehicle characteristics, including kinematics, compliance, and CG location

Suspension Engineer and Purchasing Coordinator Feb 2025 - Jun 2026

Cornell Formula SAE Racing, Cornell University, Ithaca, NY

- Designed and manufactured a custom welding fixture for 24 suspension links, targeting tolerance, weldability, simplicity
- Oversaw \$200,000+ in procurement across 10 subteams, growing the purchasing team from 2 to 7 and standardizing communication and workflows, and strengthening accountability for order tracking, records, and bookkeeping
- Served as point of contact between the team and suppliers, coordinating lead times, shipping, and unexpected issues
- Wrote a 200+ page supplier RFQ for the FSAE Business Case, translating our engineering requirements into BOMs, manufacturing specifications, quality standards, and delivery timelines

Undergraduate Teaching Assistant Dec 2025 - May 2025

Department of Physics, Cornell University, Ithaca, NY

- Lead weekly recitation for 20 students in Physics II, guiding rigorous multi-step derivations and analytical problem solving
- Strengthen physics knowledge and teaching skills through a concurrent course in evidence-based physics instruction

Seasonal Teller May 2025 - Aug 2025

Middlesex Savings Bank, Wayland, MA

- Provided in-branch and phone customer service to hundreds of clients, including resolving conflicts and frustrations
- Processed thousands of customer transactions, including deposits, withdrawals, loan payments, check cashing, and CDs
- Ensured day-to-day compliance with identity verification, dual control, audits, fraud, and documentation procedures

ADDITIONAL EXPERIENCE

Team Lead, Mathematical Modeling Team, Newton North High School, Newton, MA Nov 2021 - May 2024

Captain, Intramural Soccer Team, Cornell University Mar 2025 - Present

Tour Guide, Office of Visitor Relations, Cornell University May 2026 - Sep 2026

SKILLS & INTERESTS

Technical Skills: Microsoft Excel, Python, MATLAB, Autodesk Inventor, Fusion 360, Manual & CNC Machining

Professional Skills: Customer Service, Cross-Functional Collaboration, Project Management, Attention to Detail

Languages: Mandarin (Conversational), Japanese (Limited Proficiency)

Interests: Soccer, Real Madrid FC, Formula 1, Entrepreneurship/Product Development, Stocks, Biking, Skiing, Dragon Boating